[INCH-POUND]
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SUPERSEDING
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27 June 1990

FEDERAL SPECIFICATION

BURNERS, SINGLE: LIGHT OIL, GAS, AND LIGHT OIL-GAS COMBINATION FOR PACKAGED HEATING BOILERS (UP TO 320,000 BTU/HR THERMAL OUTPUT CAPACITY)

The General Services Administration has authorized the use of this specification by all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers single burners fired on gas, light oil, or a combination of gas and light oil designed for installation in a low pressure steam boiler or a low temperature hot water generator.

1.2 Classification. Burners will be one of the following classes as specified (see 6.2 and 6.7).

Class 2 - Light oil fired.
Class 3 - Gas fired.
Class 5 - Combination light oil and gas fired.

2. APPLICABLE DOCUMENTS

2.1 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data which may improve this document should be sent to: Commanding Officer (Code 15E2), Naval Construction Battalion Center, 1000 23rd Avenue, Port Hueneme, CA 93043-4301, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4530

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AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME CSD-1 - Controls and Safety Devices for Automatically Fired Boilers.

(Private sector and civil agencies may purchase copies of these voluntary standards from the American Society of Mechanical Engineers, United Engineering Center, 345 East 47th Street, New York, NY 10017.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 396 - Fuel Oils.

(Private sector and civil agencies may purchase copies of these voluntary standards from the American Society for Testing and Materials, 100 Barr Harbor Dr., West Conshohocken, PA 19428-2959.)

HYDRONICS INSTITUTE

Test and Rating Standard for Heating Boilers.

(Private sector and civil agencies may purchase copies of these voluntary standards from the Hydronics Institute, 35 Russo Place, P.O. Box 218, Berkeley Heights, NJ 07922.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 - National Electrical Code.

(Private sector and civil agencies may purchase copies of these voluntary standards from the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269.)

(DoD activities may obtain copies of those adopted voluntary standards listed in the DoD Index of Specifications and Standards free of charge from the Defense Automated Printing Services, Attn: DoDSSP, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The classes of burners covered by this specification shall either be furnished as part of a new packaged boiler or shall be a replacement or conversion on an existing boiler as specified (see 6.1.1, 6.2, and 6.4). The burner shall include a control system, a combustion control system, fuel piping, and control panel. All burners shall be factory-wired and assembled

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as practical except for readily installed appurtenances, such as pressure and temperature gages, oil heater piping, and wiring of the burners to the control panel. The burners shall be complete and ready for operation when installed on the applicable equipment and connected to the fuel and electrical supplies.

3.2 Codes and standards. All burners shall conform to the applicable requirements of ASME CSD-1 and NFPA 70.

3.3 Standard commercial product. The burner shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the burner being furnished. A standard commercial product is a product which has been sold or is currently being offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

3.4 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. Unless otherwise specified, none of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification.

3.5 Interchangeability. All units of the same classification furnished with similar options under a specified contract shall be identical to the extent necessary to ensure interchangeability of component parts, assemblies, accessories, and spare parts.

3.5.1 System of measurement. The dimensions used in this specification are not intended to preclude the use of the metric system of measurement in the fabrication and production of the material, individual parts, and finished product, provided form, fit, and function requirements are satisfied.

3.6 Design and construction. The burner shall be designed and constructed in accordance with the applicable requirements of 3.2 and the additional requirements specified herein.

3.6.1 Fuel characteristics. Unless otherwise specified (see 6.2), light oil fired burners shall be designed and constructed to operate on the fuel oil characteristics described in ASTM D 396. The fuel oil shall be grade 1 or grade 2 as specified (see 6.2). When oil burners are required, the fuel oil supply temperature and pressure shall be as specified (see 6.2). When gas burners are required, fuel gas supply pressure shall be as specified (see 6.2). Unless otherwise specified

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(see 6.2), gas burners shall be designed and constructed to operate on the following fuel gas characteristics:

a. Natural gas, (percent by volume, as fired).

(1) Ultimate analysis:

(a) Methane ----- 90.0
(b) Ethane ------- 5.0
(c) Nitrogen ----- 5.0
(d) Ash ---------- Nil
(e) Moisture ----- 0.01

(2) Higher heating valve - 1,075 British thermal units per cubic foot (Btu/cu ft) (40,054 kilojoule per cubic metre (kj/m^3)).

3.6.2 Installation site. The burners shall be designed and constructed to operate for the following installation site characteristics as specified (see 6.2):

a. Site elevation in feet (metre) above sea level.

b. Anticipated maximum, minimum, and average ambient air temperature in degrees Fahrenheit (°F) (degrees Celsius (°C)).

3.6.3 Oil fired burners. When oil fired burners are required, the burners shall produce a clean, stable flame at all firing rates using the described grade of fuel oil as specified in 3.6.1. The burner flame shape shall fit the furnace without impingement on or overheating of the furnace walls. Burner parts shall be protected from overheating. The design of the fuel oil system including safety interlocks, limit switches, control valves, and fuel oil piping shall be in accordance with ASME CSD-1. Fuel oil piping shall withstand 1.5 times the maximum operating pressure of the oil supply system without leakage. The fuel oil system shall include an atomization system, fuel oil pump, strainer, and relief.

3.6.4 Gas fired burners. When gas fired burners are required, the burners shall produce a clean, stable flame at all firing rates using the described gas fuel. The burner flame shape shall fit the furnace without impingement on or overheating of the furnace walls. Burner parts shall be protected from overheating. The design of the system including safety interlocks, limit switches, control valves, and fuel gas piping for burners shall be in accordance with ASME CSD-1. Fuel gas piping shall withstand 1.5 times the maximum operating pressure of the fuel gas supply system without leakage. Partial premix burners shall be provided with an effective design to protect flame propagation back into the mixing chamber or gas and air supply lines.

3.6.5 Combination gas and oil fired burners. When combination oil and gas burners are required, the burners shall consist of an oil burner and a gas burner each conforming to the requirements of 3.6.3 and 3.6.4, respectively. The two burners shall be constructed as an integral

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combination burner suitable for firing either fuel separately. Provisions shall be incorporated for withdrawing, shielding, or otherwise preventing the burner from plugging while operated on either fuel for extended periods. Except as otherwise specified herein, the fuel systems and fuel safety interlocks shall consist of separate systems. A fuel change shall require a shutdown to initiate. The burner shall be so designed that the flame failure, combustion controls, fuel cutoff limit switches and other applicable controls will be transferred from one fuel to the other by a single manual selector switch.

3.6.6 Control system. Each burner shall be provided with a control system in accordance with the applicable requirements of ASME CSD-1.

3.6.6.1 Safety interlocks and limit switches. Each control shall have provisions for the connection of any safety interlock switches or limit controls that are required on the boiler to satisfy the applicable requirements of ASME CSD-1. All nonrecycling shutdowns shall require a manual reset of the control system. All safety interlocks and limit switches, including boiler and equipment limit switches, shall be wired on the hot side of the burner limit control circuit and not in any external circuits such as the pilot or fuel valve circuits.

3.6.7 On-off combustion control systems. The on-off combustion control system shall automatically regulate the firing cycle in accordance with the local demand. The combustion control system shall maintain an air-to-fuel mixture within the limits required for stable and efficient combustion. The continuous maximum firing rate shall be as required to maintain the maximum boiler thermal output capacity required in 3.7 ± 10 percent of full scale. The percentage tolerances shall not be applied in a manner that would permit the high-fire input of the burner to be increased beyond the maximum rated input marked on the boiler assembly. The fixed rate, on-off control system shall operate in response to load demand to cycle the burner on and off. A fixed relationship shall exist between the volume of air and fuel admitted to the burner. The air flow control damper shall be manually adjustable to permit changing of the air-to-fuel ratio when required by firing conditions. After the main flame is proven, units with on-off controls shall continue to fire until the load demand is satisfied.

3.6.8 Electrical requirements. Electrical wiring design practices shall be in accordance with the applicable requirements of NFPA 70. All control circuits shall be two-wire, one side grounded, not greater than nominal 120 volts, and be protected with suitable fuses or circuit breakers.

3.6.9 Combustion detector. The combustion detector shall be installed so as to not respond to igniter sparks or hot refractory. The combustion detector for igniter flames shall detect only flames of sufficient intensity to safety light the main burner.

3.7 Performance. The thermal performance of the burner shall be based upon the ability of the boiler to produce the required thermal output when the burner is fired using the fuel specified in 3.6.1. The maximum thermal output requirement of the packaged boiler that the burner is to be installed in shall be as specified (see 6.2). The performance of the boiler will be evaluated in accordance with the Hydronics Institute Testing and Rating Standard for Heating Boilers. Operation of the burner equipment shall be performed to the requirements of this specification.

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3.8 Environmental requirements. The emission requirements shall be met at the maximum required continuous output. The burner shall meet environmental rules and regulations. Possible emission requirements to be considered are opacity, particulate, oxides or nitrogen, sulfur dioxide, and carbon monoxide. Other emission requirements may be imposed.

3.9 Cleaning, treatment, and painting. Bare steel surfaces shall be cleaned and a primer coat applied. Paint shall be applied to normally painted surfaces in good commercial practice.

3.10 Identification marking. Identification shall be permanently and legibly marked directly on the burner or on a corrosion-resistant metal plate. Identification shall include manufacturer's name, burner model and serial number, and year of manufacture. When specified (see 6.2), the Department of Defense contract number that this equipment is procured under shall also be marked on the burner or identification plate.

3.11 Workmanship. Workmanship shall be of such quality as to produce burners in accordance with the ASME Code, as applicable and shall meet the requirements of this specification and standards prevailing among manufacturers who normally produce this type of equipment.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification, where such inspections are deemed necessary to ensure that supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in this document shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

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a. Quality conformance inspection (see 4.2.1).
b. On-site inspection (see 4.2.2).

4.2.1 Quality conformance inspection. This inspection shall be performed at the manufacturer's facilities. Unless otherwise specified (see 6.2), the quality conformance inspection shall include the examination of 4.3, and the fuel system pressure test of 4.4.

4.2.2 On-site inspection. The on-site inspection shall be in addition to the quality conformance inspection performed at the manufacturer's facilities. Unless otherwise specified (see 6.2), on-site inspection shall consist of the test of 4.5. All or part of the on-site inspection may be performed at the manufacturer's facilities if specified by the contracting officer (see 6.2).

4.3 Examination. Each burner and related equipment supplied under this specification shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability.

4.4 Fuel system pressure test. The preassembled fuel system piping furnished integrally with the burner shall be checked for leakage. The fuel system piping shall be hydrostatically tested for 30 minutes at 1.5 times the maximum operating pressure without leakage.

4.5 On-site test. Operational fire test and auxiliary equipment tests shall be performed on each installed burner during the performance testing of the boiler it is installed on. The performance of the boiler shall be evaluated in accordance with the Hydronics Institute Testing and Rating Standard for Heating Boilers. These tests shall be in addition to any manufacturer's standard fire test procedure, unless compliance evidence provided by the burner manufacturer is accepted at the discretion of the contacting officer or his authorized representative. Acceptable fire test results from identical or similar burner equipment installed at the same site may be used at the discretion of the contracting officer as proof that equipment procured with this specification is in compliance with one or more of the test of 4.5.1 through 4.5.8.

4.5.1 Operational fire tests. Each burner shall demonstrate control and operational conformance to the requirements of this specification under varying load conditions including the maximum required capacity. Burner capacity shall be adequate if the boiler is installed and is able to meet the maximum thermal capacity and efficiency requirements. Oil fired burners shall be tested for the ability to cold start at the specified oil temperature (see 3.6.1) in accordance with the manufacturer's instructions. Burners that exhibit (1) excessive or unexplained loss of ignition, (2) nuisance shutdown due to faulty burner of control operation, (3) improper flame, (4) excessive carbon deposits, or (5) necessity for frequent or difficult adjustments shall be rejected.

4.5.2 Sequencing. The burner shall start, operate, and stop in strict accordance with the operating sequence of the applicable requirements of ASME CSD-1.

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4.5.3 Flame failure. The operation of the flame failure control shall be verified by simulated flame and ignition failures. The igniter establishing period, the main-burner establishing period, flame-failure reaction time, and the safety-shutoff valve closing times shall be verified by stop watch.

4.5.4 Hot refractory test of combustion safeguard system. The burner shall be operated at high fire until the furnace attains steady state condition. The main fuel valve shall then be closed manually. The combustion safeguard shall drop out causing the safety shutoff valves to close within the specified flame-failure reaction time.

4.5.5 Minimum igniter test of combustion safeguard system. The fuel supply to the igniter flame shall be gradually reduced to the point where the combustion safeguard begins to drop out (sense “no flame”) but holds in until the main fuel valve opens. At this point of reduced igniter fuel supply, the igniter flame shall be capable of safely igniting the main burner. If a small igniter flame can be maintained that does not activate the combustion safeguard system but is insufficient to safely light the main burner, the burner shall be rejected.

4.5.6 Spark rejection test. With the fuel supply to both the igniter and main frame closed, the ignition cycle shall be initiated. When the spark ignition comes on, the flame detector shall not be activated. If the flame detector can activated, the burner shall be rejected.

4.5.7 Combustion controls. The accuracy range and smoothness of operation of the combustion controls shall be demonstrated by varying the load demand by varying the load to require on-off cycling. The control accuracy shall be as specified in 3.6.7. The air-to-fuel ratio controller shall be tested by attempting to create a fuel rich operating condition below the manufacturer’s pre-set limit.

4.5.8 Equipment tests. Fuel pumps, electric motors, and other burner related equipment shall be operated as part of the operational fire test. The operation of this equipment shall be closely observed during the operational tests for possible defects or nonconformance. The action of mechanical devices shall be smooth without backlash.

5. PACKAGING

5.1 Packaging requirements. The preservation, packing, and marking shall be as specified in the contract or order.

6. NOTES

(This section contains information of general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The burners covered by this specification are intended for use in packaged low pressure steam boilers or low temperature hot water generators having a thermal output capacity between 40,000 and 320,000 Btu/hr (11 722 and 93 783 watts (W)). The burner class should be

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determined in accordance with the availability or economics of the fuel supply and the intended application. Multi-fuel burners should be specified in order to ensure economic and reliable operation within the constraints of fuel availability and applicable environmental regulations

6.1.1 Selection factors. Selection factors and the best choice of options will vary depending on the size of the boiler that the burner is to be installed in. A spare burner should not be procured without knowing the specific boiler that it is intended to be used in. Variable fuel compositions and boiler configuration information needed when specifying a burner shall be known. The following items should be considered in the selection of the burner and the exercise of options herein:

a. This specification can be used for replacement or conversion installations to specify the procurement requirements for the burner. However, it should remain the responsibility of the contractor who is performing the boiler modifications to make the retrofitted boiler operate satisfactorily (see 3.1).

b. Fungus resistant varnish conforming to MIL-V-173 should be used to coat electrical components and circuit element, including terminal and circuit connections, when the burner is to be installed in humid conditions. Components and elements inherently inert to fungi or in hermetically sealed enclosures or current-carrying contact surfaces should not be coated.

c. Electromagnetic interference suppression (EMI), when required, should conform to the EMI suppression requirements and test limits for Class C3, Group I equipment as specified in MIL-STD-461. The burner should be subjected to test to determine conformance in accordance with MIL-STD-462.

6.1.2 Other fuels. The performance requirements specified herein may be based on other fuels providing the quantity on-site is sufficient for testing specified in section 4, and a guaranteed ultimate analysis and higher heat content are made available at the time of testing the burners. Corrections between the values of the analysis in section 3 of this specification and the guaranteed analysis and heating value should be made in accordance with the Hydronics Institute Testing and Rating Standard for Heating Boilers. Referenced below are some of the most commonly used fuels with their average specific gravity and higher heat content:

Fuel	Average specific gravity	Average higher heat content	
Natural gas	0.65	1,075 Btu/cu ft	(40 054 kj/m^3)
Manufactured gases	0.38	535 Btu/cu ft	(19 934 kj/m^3)
Mixed gas	0.50	800 Btu/cu ft	(29 807 kj/m^3)
Liquefied petroleum gas	1.55	2,500 Btu/cu ft	(93 148 kj/m^3)

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6.2 Acquisition documents. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Class and style of burner required (see 1.2).
c. Whether the burner is part of a new packaged boiler or a replacement or conversion on an existing boiler (see 3.1).
d. Fuel oil characteristics for light oil fuel, if other than as specified (see 3.6.1).
e. Grade of fuel to be used (see 3.6.1).
f. Fuel oil supply temperature and pressure for oil fired burners (see 3.6.1).
g. Fuel gas supply pressure, when gas is fired burners are required (see 3.6.1).
h. Characteristics of fuel gas, if other than as specified (see 3.6.1).
i. Installation site elevation characteristics (see 3.6.2):

    (1) Site elevation in feet above sea level.
    (2) Anticipated maximum, minimum, and average ambient air temperature (°F).

j. Maximum thermal output capacity of the boiler that the burner is to be installed in (see 3.7).
k. When the Department of Defense contract number is to be marked on burner or identification plate (see 3.10).
l. Quality conformance inspection, if other than as specified (see 4.2.1).
m. On-site inspection, if other than as specified (see 4.2.2).
n. When all or part of on-site inspection is to be performed at the factory (see 4.2.2).

6.3 Compliance. Prior to approval of shipment, the contractor should submit to the contracting officer or authorized representative satisfactory evidence that the proposed burner to be furnished under this specification meets the requirements of the ASME CSD-1 and NFPA 70.

6.3.1 ASME CSD-1 requirements. Acceptable evidence of meeting the requirements of the ASME CSD-1 should be the receipt of the manufacturer's certificate of compliance (see 6.3), starting the equipment procured with this specification is in accordance with the applicable provisions of ASME CSD-1.

6.3.2 NFPA requirements. Acceptable evidence of meeting the requirements of NFPA 70 should be the manufacturer's certificate of compliance (see 6.3) with the applicable requirements of NFPA 70.

6.3.3 Technical manuals. The manufacturer should provide two commercial technical manuals normally prepared and supplied with the boiler. Technical manuals should specify a parts list and part number for the equipment issued.

6.3.4 Burner tips. The contractor should provide an extra burner tip for each burner based on the fuel to be fixed.

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6.4 Definitions. Definitions used in this specification have been in accordance with the ABMA Lexicon of Boiler and Auxiliary Equipment. Other terms used in this specification should be interpreted in accordance with the referenced NFPA and UL standards.

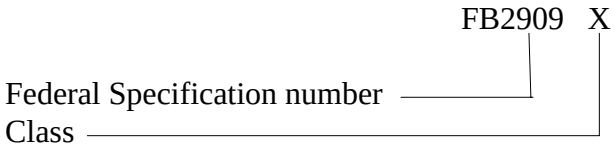
6.4.1 Replacement. A burner replacement is defined in this specification as the simple replacement of existing equipment without changing the fuel, thermal heat duty, or air pollution emissions.

6.4.2 Conversion. A burner conversion is defined in this specification as the replacement of existing equipment to change the type of fuel used, the control system, the thermal heat duty, the boiler air pollution emission or other significant operating variable.

6.5 Supersession data. This specification replaces Military Specification MIL-B-18797G, dated 27 June 1990.

6.6 Classification cross reference. Classification used in these specification (see 1.2) are identical to those found in the superseded Military Specification MIL-B-18797G.

6.7 Part or Identifying Numbers (PINs). The specification number and class are combined to form PINs for burners covered by this document (see 1.2). PINs for the burners are established as follows:



6.7.1 Class. The class of units (see 1.2) is identified by a single numerical character (see table I).

TABLE I. Code number to class.

Class	Code
II	1
III	2
V	3

6.8 Subject term (key word) listing.

Combination oil and gas fired
Gas fired
Oil fired

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MILITARY ACTIVITIES:

Custodians:
Army - CR4
Navy - YD1
Air Force - 99

Review Activities:
Navy - MC
Air Force - 84
DLA - CC

CIVIL AGENCY COORDINATING ACTIVITY:

GSA-FSS

Preparing Activity:
Navy - YD1

(Project 4530-0008)

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